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An evaluation of catchment-scale phosphorus mitigation using load apportionment modelling

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ABSTRACT

Functional relationships between phosphorus (P) discharge and concentration mechanisms were explored using a load apportionment model (LAM) developed for use in a freshwater catchment in Ireland with fourteen years of data (1995–2008). The aim of model conceptualisation was to infer changes in point and diffuse sources from catchment P loading during P mitigation, based upon a dataset comprising geospatial and water quality data from a 256 km^2 lake catchment in an intensively farmed drumlin region of the midlands of Ireland. The model was calibrated using river total P (TP), molybdate reactive P (MRP) and runoff data from seven subcatchments. Temporal and spatial heterogeneity of P sources existed within and between subcatchments; these were attributed to differences in agricultural intensity, soil type and anthropogenically-sourced effluent P loading. Catchment rivers were sensitive to flow regime, which can result in eutrophication of rivers during summer and lake enrichment from frequent flood events. For one sewage impacted river, the LAM estimated that point sourced P contributed up to of 90% of annual MRP load delivered during a hydrological year and in this river point P sources dominated flows up to 92% of days. In the other rivers, despite diffuse P forming a majority of the annual P exports, point sources of P dominated flows for up to 64% of a hydrological year. The calibrated model demonstrated that lower P export rates followed specific P mitigation measures. The LAM estimated up to 80% decreases in point MRP load after enhanced P removal at waste water treatments plants in urban subcatchments and the implementation of septic tank and agricultural bye-laws in rural subcatchments. The LAM approach provides a way to assess the long-term effectiveness of further measures to reduce P loadings in EU (International) River Basin Districts and subcatchments.

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1. Introduction

Freshwater eutrophication caused by phosphorus (P) enrichment is a global issue: the number of waterbodies displaying extensive P pollution events has increased substantially over the last decade (Hilton et al., 2006; Seitzinger et al., 2010). European Water Framework Directive (WFD) requires member states to achieve good status in lakes and rivers by 2015 (European Commission, 2000) and those susceptible to the affects of high P transfers need to be identified in order to ensure the cost-effective implementation of P mitigation policies. As P-related water quality problems require a catchment-based approach, there is a need to understand the delivery pathways, mechanisms and impacts of catchment management policies on the sources of P entering rivers and lakes (Neal and Heathwaite, 2005; Neal and Jarvie, 2005).

Natural sources of P (e.g., atmospheric deposition, natural weathering of soil, and detritus from riparian vegetation) rarely cause ecological problems, whereas ecosystem effects from excessive inputs of anthropogenic P, from point or diffuse sources, can be profound (Pieterse et al., 2003; Jarvie et al., 2006; Liu and Chen, 2008). Point sources include wastewater treatment plants (WWTPs), septic tanks with direct connection to rivers, farmyards and industrial activities; diffuse sources comprise runoff from urban areas, septic tank soak-aways, forestry and agriculture (Withers and Jarvie, 2008). The transfer pathways of P from both sources to waterbodies differ and have contrasting relationships with flow conditions (Chapman, 1996). Point sources of P deliver relatively constant loads during the hydrological cycle of a river and are independent of flow. Conversely, diffuse sources are highly dependent on flow for transfer from land to water (Edwards and Withers, 2008); an increase in P concentration with flow is caused by a combination of nutrients washed off from the land in runoff, river bed remobilisation and erosion processes (Best et al., 1997). During flood events, the diffuse sources dominate instantaneous river loads (Webb et al., 1997; Tunney et al., 2000; Jordan et al., 2005a). In the intervals between high flow events, P concentrations are low when runoff pathways are disconnected and/or when point sources are absent or a sufficient dilution factor exists. Phosphorus concentrations can be high at low flow when a

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point source is operating even if the instantaneous river loads are low in comparison to those that occur during flood events (Arnscheidt et al., 2007; Douglas et al., 2007). Such high P concentrations with low flows in summer can be more influential in maintaining a eutrophic status than the annual contribution of point sources would indicate (Bowes et al., 2010).

Mobilisation of P from a specific source varies spatially and temporally (Donohue et al., 2005). Exports of P from a catchment are primarily a function of landuse (Wood et al., 2005), soil conditions (Jordan et al., 2005a), the connectivity between hydrology and catchment management (Foy et al., 1995; Foy and Lennox, 2000; Ulén and Jakobsson, 2005) and P loading (Donohue et al., 2006). A variety of approaches have been applied to determine the sources and transfer dynamics of P loads (May et al., 2001; Bowes et al., 2005a; Withers and Jarvie, 2008). The source approach estimates P loads before entry into a watercourse, using data from the monitoring of point contributors (Foy et al., 1995), although measuring diffuse transfers using this method can be difficult. In the P export coefficient approach, coefficients are applied in order to estimate P loading using catchment attributes, such as population density or the extent of different land covers and land uses (Johnes, 1996). Both of these methods require detailed information on catchment activities and physical attributes but they are unable to accommodate random pollution events or the effects of annual hydroclimatic variations. Using an approach based on water quality monitoring to calculate P load (Quilbé et al., 2006) can also be problematic, because such an approach does not usually differentiate between sources. This problem can be addressed, however, through the development and application of a load apportionment model (LAM), which provides a means of separating contributions from point and diffuse loads by examining the relationship between P concentration and river runoff (Bowes et al., 2008, 2009, 2010; Howden et al., 2009; Jarvie et al., 2010).

Determining the effectiveness of P mitigation management programs is important for policy evaluation (Cherry et al., 2008). However, relatively few studies have investigated the affects of P mitigation on P sources at a catchment scale during long-term policy implementation, owing to the resources required for high-temporal resolution monitoring (Foy et al., 1995; Jarvie et al., 2002; Kronvang et al., 2005, 2008; Bechmann et al., 2008; Rao et al., 2009). In examining P dynamics in the context of a Programme of Measures (POMs) aimed at mitigating P losses in a nutrient sensitive catchment, the current paper seeks both to test the LAM approach and to provide a basis for policy evaluation. The study catchment contains a lake, Lough Sheelin, which has varied in trophic status in recent years. A LAM was developed to describe observed relationships between P and discharge data to aid the assessment of catchment response to the implementation of P-related POMs.

2. Material and methods

2.1. Study site

Lough Sheelin (18.1 km^2) (53°08′ N, 7°33′ W, 65.4 amsl) is a shallow (mean depth 4.5 m) lake, with a catchment area of 256 km^2, located in the drumlin belt of central-north Ireland. The catchment has both agricultural and urban sources of P. Ten tributaries drain into the lake, which has a single outflow via the River Inny, located in the southwest end of the lake. Total phosphorus (TP) and molybdate reactive phosphorus (MRP) concentration data collected from 1982 to 2008 for the study catchment were provided by Inland Fisheries Ireland (IFI). The flow data were provided by the Environmental Protection Agency (EPA) Ireland and IFI. The research that underpins this paper focuses on the seven tributaries that had the most complete dataset (Fig. 1 and Table 1).

A major geologic strike divides the Sheelin catchment into a northern, upland part that is predominately underlain by carboniferous limestone, and a southern part characterised by shale and sandstone. The subcatchments Halfcarton, Mountnugent and Ross are characterised by high relief and steep slopes, whereas the topography in the remainder of the catchment has lower altitude. As is typical for the drumlin belt in Ireland, soils generally have low permeability and are vulnerable to water-logging and nutrient loss via surface and near-surface hydrological pathways (Kerins et al., 2007). Landuse in the catchment is dominated by managed grasslands, with cattle and pig farming representing the main agricultural enterprises. Two small towns, Ballyjamesduff (human population 1,700 as of 2006) and Oldcastle (human population 1300 as of 2006) (CSO, 2006), are located in the Mountnugent and Ross subcatchments respectively. Each town has a WWTP. Effluent from a meat processing plant, located close to Ballyjamesduff, discharges into the Mountnugent river. Human settlement in the remainder of the catchment is characterised by scattered dwellings, each linked to domestic septic systems for waste water treatment (Keys and Gibbons, 2008). Between 1996 and 2006 the populations of Ballyjamesduff and Oldcastle increased by 60% and 40%, respectively, while rural population densities also increased, ranging from 10% in Carrick to 53% in Mountnugent (CSO, 2006).

Several phases of catchment-scale P mitigation have been implemented since 1999. An upgrade and expansion of the Ballyjamesduff WWTP from primary to tertiary chemical treatment and the issue of an integrated pollution prevention control (IPPC) license to the meat processing plant both occurred in 1999 (Kerins et al., 2007). An interim P removal regime was introduced at the overloaded Oldcastle WWTP in 2003. Catchment-wide, septic tank bye-laws were officially introduced in 2004, following small improvements initiated in 2000. Agricultural bye-laws to regulate the storage and management of livestock wastes were in place by 2003 (Keys and Gibbons, 2006); these have been superseded by national regulation under the Nitrates Directive. Licensing of trade discharges to water, septic tank compliance, pollutant investigations and farm surveys were also introduced in 2003.

2.2. Data sources and structure

Total P and MRP (μg l^{-1}) concentrations for 1995 to 2008 were available from seven monitoring stations in the study catchment. River samples were collected three times per week throughout the monitored period. Flow rates for the seven rivers were collected as daily mean flows (m^3 s^{-1}) from 15 min measurements of water level and rated staff gauges (m). TP was determined by persulphate digestion (Eisenreich et al. 1975) and MRP was measured using the colorimetric acid molybdate method (Murphy and Riley, 1962).

Censored P concentrations (values recorded as below detection limit) were replaced with one half of the detection limit e.g. <6 μg l^{-1} was replaced with 3 μg l^{-1} (Qian et al., 2007; Reimann et al., 2008). A full annual discharge record was created by estimating values for missing or omitted flows on the basis of strong correlations (R^2>0.8) with yearly flow values for neighbouring rivers with a continuous flow record (Mwakalila, 2003). Flow values for each river were normalised per unit area by calculating the surface runoff depth (mm day^{-1}) (Lennox et al., 1997; Jordan et al., 2005b). The calculation of runoff depth enabled direct comparison of water drainage between the subcatchments. The P loads estimated from modelling the relationships between P and runoff by LAM application were also normalised by subcatchment area. Data on P export per unit area were thus provided and disproportionate fluxes of P from subcatchments highlighted. Surface flow is the main delivery pathway to the rivers in the catchment (Kerins et al., 2007). Runoff between and within subcatchments was assumed to be spatially uniform and influenced by the same catchment characteristics and processes.

Phosphorus and flow data were paired according to date and organised in a database by hydrological year (1st October to 30th September). The database was divided into seven periods (e.g. 1995–1996 ranges from 1st October 1994 to 30th September 1996).

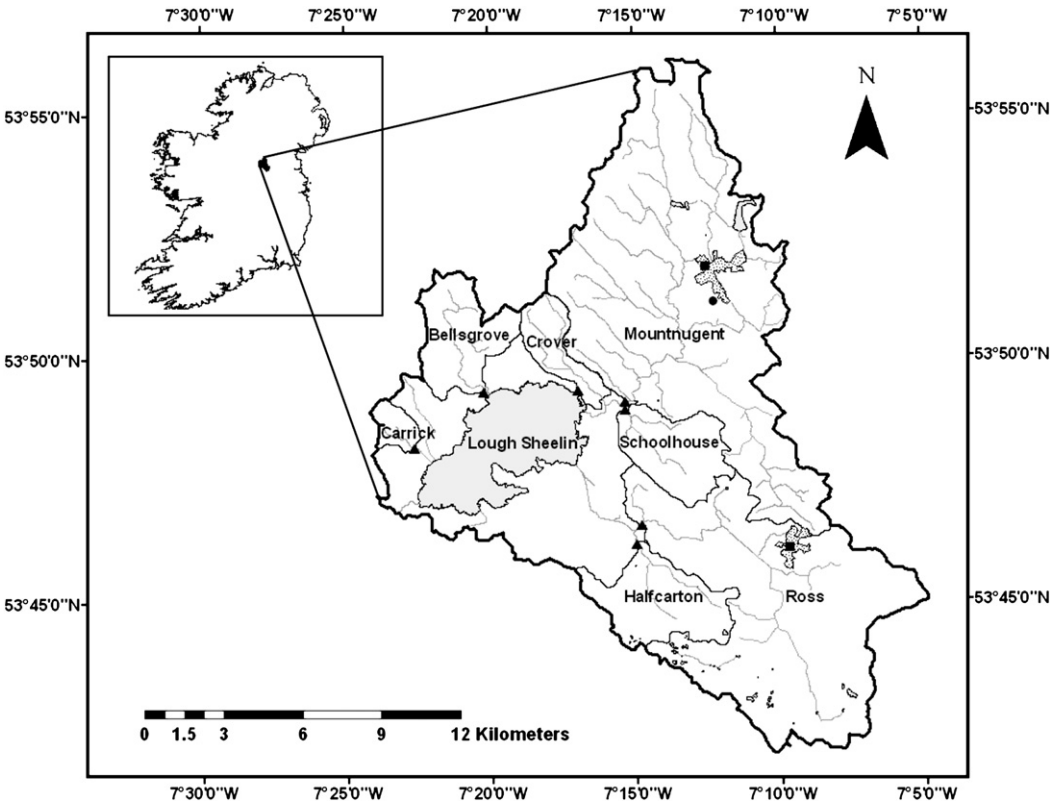


Fig. 1. The Lough Sheelin catchment showing its location in Ireland (inset) and the seven study subcatchments. Water quality sample site = ▲, Industrial plant = ●, waste water treatment plant (WWTP) = ■, river = ——, lake = ▭, urban area = ▨.

Phosphorus concentration and runoff data were log-transformed to achieve a normal distribution and to minimise the affect of outliers.

GIS software and extensions (Maidment, 2002; Ormsby et al., 2008) were used to manipulate spatial data. Subcatchment boundaries were delineated from a digital elevation model and river vector data by using each monitoring site as a pour–point. Data on flow, geology, land cover, soils, human population and cattle stocking density statistics were extracted using the subcatchment boundary outlines. An estimation of human population numbers between census years for each subcatchment was calculated from linear interpolation of census data (CSO, 2006).

2.3. Model development

Three concentration-discharge relationships are commonly recognised for nutrients in rivers (Fig. 2): point sources (a — inverse curve),

Table 1
Descriptive attributes of the seven study subcatchments in the Lough Sheelin catchment.

Subcatchment	Area (km^2)	Irish grid reference	Strahler stream order	Q5:Q95 ratios	Soil type		Land use [b]	Human population (1998 and 2008)	P point source
					P desorption Index [a]	Soil type			
Bellsgrove	11.6	N 440 862	3	63.2	2.8	Acid brown earths, podzolics, gleys	Permanent pasture 50% farmed 51% improved	186–249	Septic tanks
Carrick	2.4	N 419 838	2	101.8	3.1	Acid brown earths, podzolics, gleys	Permanent pasture 60% farmed 35% improved	39–42	Septic tanks
Crover	6.1	N 471 859	2	39.6	3.0	Gleys, brown earths, podzolics	Permanent pasture 62% farmed 53% improved	105–125	Septic tanks
Halfcarton	14.3	N 480 797	2	28.8	1.8	Brown earths, grey brown podzolics , rezinas, lithosols	Permanent pasture 58% farmed 35% improved	204–246	Septic tanks
Mountnugent	91.6	N 489 857	4	30.2	2.7	Gleys, peatland	Permanent pasture 78% farmed 36% improved	2884–5143	1. WWTP for 35% population 2. Industrial P source 3. Septic tanks
Ross	63.1	N 487 806	3	17.2	1.8	Brown earths, grey brown podzolics	Permanent pasture 71% farmed 30% improved	1782–2232	1. WWTP for 50% population 2. Septic tanks
Schoolhouse	10.9	N 480 836	2	98.0	1.9	Brown earths, peatland, podzolics	Permanent pasture 84% farmed 51% improved	247–329	Septic tanks

[a] The potential for P desorption from soil (Daly and Mills, 2006). A high value indicates an increased risk of P loss by desorption.
[b] CORINE (CLC,2000).

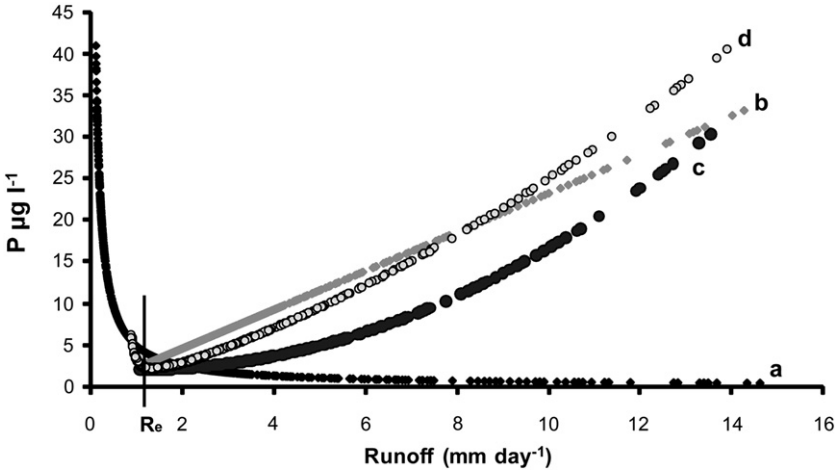


Fig. 2. Relationships between phosphorus concentration (P µg l^{-1}) and discharge (mm day^{-1}), (a) point source – inverse relationship (b) diffuse source – linear relationship (c) diffuse source – exponential relationship (d) combined point and diffuse relationships – U shaped relationship and the load apportionment model (d) showing R_e.

diffuse sources (b and c — linear or exponential curve) and a third (d — u-shaped curve) describing a situation where contributions from both point and diffuse sources occur (Mohaupt, 1986; Chapman, 1996; Edwards and Withers, 2007). In the latter scenario, when both point and diffuse nutrient sources contribute to river loading, nutrient dilution is the critical factor at a low discharge range, but at higher flows dilution is suppressed and diffuse P has the largest influence on loading. The R_e parameter provides an estimation of the runoff depth at which point sources (runoff$<R_e$) or diffuse sources (runoff$>R_e$) are the predominant influence on the river expressed as a temporal proportion of flows in any hydrological year (Bowes et al., 2008).

As general linear modelling (GLM) indicated a seasonality trend in the data (Ho, 2006; Field, 2009), the database was split into S1 (November to May) and S2 (June to October) periods. Several additive models combining the relationships between P concentration as a function of runoff were tested by calibration to the study database using non-linear least square regression in PRISM 5 Graphpad software (Motulsky, 2007). The model that most accurately described the behaviour of the data and provided realistic outputs was selected and is described in Eq. 1;

$$Cp = a^*\left(\frac{1}{Q}\right) + b^*(Q) + c^*\left(Q^2\right) \quad [1]$$

where Cp is empirical P (µg l^{-1}), Q is empirical runoff depth (mm day^{-1}) and a, b and c are estimated model coefficients. The model components describe the relationship of P with runoff along a hydrograph: point sourced P showing an inverse relationship with runoff (a), diffuse sourced P showing a linear (b) and exponential (c) relationship with runoff (Fig. 2). The assumptions underlying the model consist of the dilution of point sources, flow dependence of diffuse contributions and a calibration database with a temporal resolution that captures representative samples. The 'a' coefficient estimates P from continuous sources. Diffuse sources represented by the 'b' coefficient are dependent on moderate runoff. The 'c' coefficient represents exponential increases in concentration owing to sheet erosion during storm events. The model coefficients were not constrained to a numerical range prior to analysis in order to permit consideration of concentration-discharge hysteresis. Uncertainties in estimations were quantified by confidence intervals ($p<0.05$). Individual models were derived for each season type (S1 and S2), hydrological year combination (e.g. 1995–1996 etc) and river monitoring site.

The 'a' coefficients (P g km^{-2} day^{-1}) estimated from Eq. (1) were converted to annual (S1 plus S2) point P loading (kg y^{-1}) using Eq. (2);

$$P_{point} = \frac{a^*365.25^*CA}{1000} \quad [2]$$

where P_{Point} = annual point P loading (P kg y^{-1}), a = model coefficient (P g $km^{-2}day^{-1}$), CA = subcatchment area (km^2). R_e values were calculated by dividing the estimated 'a' coefficient over the sum of the estimated 'b' and 'c' coefficients.

Empirical daily runoff data were applied to each estimated seasonal model to derive the annual total P (point and diffuse) load (L_f, kg yr^{-1}) delivered to each river. L_f was used to quantify the percentage of P derived from point and diffuse sources and was calculated through a two stage process. First, individual models for each season and time period were fitted using empirical daily runoff data (Q) and the model coefficients a, b and c (P load from each delivery source estimated from Eq. (1)) to predict a daily P concentration when no samples were taken (Cp_p,) using Eq. (3);

$$Cp_p = a^*\left(\frac{1}{Q}\right) + b^*(Q) + c^*\left(Q^2\right) \quad [3]$$

Second, estimated (Cp_p) values from Eq. (3) were incorporated with the empirical daily P values (Cp) to construct a complete database of daily P concentrations (Cp_d). Values of (Cp_d) were multiplied by daily runoff (Q) to give daily P loading, which is summed to calculate annual total P loading in Eq. 4;

$$L_f = \sum_{j=1}^{n=365.25} Cp_d{}^*Q \quad [4]$$

where Q is empirical daily runoff (mm day^{-1}), Cp_d is daily P concentration and L_f represents total P load per unit area (kg km^{-2} yr^{-1}) for a hydrological year (n = 365.25).

Model accuracy was tested through a 10-fold cross validation method using 28 calibrated MRP models (two seasonal models for each of the seven rivers). A randomised subset of the data (10%) was partitioned into a validation subset (VS). The remaining 90% of the data, the estimation subset (EV), were used to estimate model coefficients from Eq. (1). The predicted MRP values were compared with the empirical VS data. The success of model prediction was established by the percentage of MRP values estimated accurately in the VS and expressed as the square root of the sum of squared differences (RMSE) between empirical and predicted values divided by the number of data points in VS. The RMSE was divided by an average MRP value in each VS to compare results of each subcatchment (CV-RMSE).

2.4. Alternative methods of load estimation

Although septic tanks are usually considered potentially important diffuse sources of nutrients (Hanrahan et al., 2001; Drolc and Zagorc

Koncan, 2002) they are also known to operate as an accumulation of small and constant point sources (Arnscheidt et al., 2007). Consequently, export coefficients (E_i) for MRP transfer from septic tanks were used in the current research as a basis for approximating catchment point MRP load. The MRP loadings from septic tanks in rural areas for each of the subcatchments were calculated as the product of their resident populations and an E_i of 0.26 kg MRP $cap^{-1}y^{-1}$ (LLCMMS, 2000; Carvalho et al., 2005). A 30% overall improvement of septic tank efficiency was anticipated when the regulatory controls on septic tanks were officially implemented in 2004 (Keys, 2008). This reduction factor was incorporated into the estimated export coefficient MRP point loads.

Molybdate reactive P concentrations in discharges from large point sources in the Mountnugent and Ross subcatchments for 1999 onwards were calculated from averaged values from effluent discharge records (flow m^3 day^{-1} and MRP mg l^{-1}). Waste water treatment plant MRP loadings prior to 1999 were estimated as the product of the town population and an export coefficient of 0.66 kg P cap^{-1} y^{-1} (Azzellino et al., 2003), which included the soluble P removal rates at WWTPs of 7.5% at Ballyjamesduff WWTP and 30% at Oldcastle WWTP (Carvalho et al., 2005). Long term monitoring records were not available prior to the IPPC licensing of the meat processing plant. An average effluent volume of 10.12 m^3 h^{-1} and an MRP concentration of 1.85 mg l^{-1}, as described by Cooney (2006), were therefore used.

Annual flow-weighted P loads, using the product of all empirical concentration-discharge pairs, divided by total discharge, were calculated for comparison with total LAM loads (diffuse plus point) calculated from Eq. (4) for both P fractions (Salles et al., 2008).

3. Results

Fig. 3 provides an example of time series plots of empirical MRP and runoff data for the Mountnugent and Bellsgrove rivers (additional plots for the other subcatchments are shown in Fig. S1 of the Supplementary material). The data indicated a declining trend in MRP in all subcatchments during 1995–2008. Seasonal Kendall trend tests (Helsel and Hirsh, 1991) on the hydrograph data did not reveal any significant long-term shifts in runoff values that could have influenced P trends. The largest reduction in empirical MRP was in Mountnugent; the smallest was in Halfcarton. A distinct seasonal cycle of P concentration and runoff values was evident in the dataset: higher P concentrations in the summer to autumn months, corresponding with lowest river runoff, highlighted the influence of point sources. Estimations of TP and MRP model coefficients for S1 (November to May) were significantly different ($p<0.01$) from S2 (June to October) coefficients, according to GLM analysis, and both S1 and S2 components were therefore included in the LAM.

Using the LAM methodology on the full, seasonally adjusted dataset generated 195 individual model and 585 model coefficient estimates of MRP and TP loads. The models showed a typical, intermediate concentration-runoff relationship (u-shaped, Fig. 4) indicating influences by both point and diffuse P transfer across a flow gradient. Seasonal models were fitted for all cases except Crover MRP (2005–2006) in which full biennial values were used because the estimated point source winter coefficient was negative. All point source coefficients were significantly different from zero except for Bellsgrove MRP (2005–2006, S1), Schoolhouse MRP (2003–2004, S1 and S2) and Schoolhouse TP (2003–2004, S1). A negative 'c' coefficient, evident in several models (Fig. 4d), was likely a result of hydrochemical hysteresis. The percentage of times that the river load was dominated by point source (runoff$<R_e$) was also calculated for each seasonal model. Model estimates of annual TP and MRP loads (point and diffuse sources) were standardised for catchment area (km^2) and the dominance in time (%) was expressed for each source, fraction and subcatchment (Table 2).

On the basis of the 10% subset, winter models had a higher percentage of validation and lower CV-RMSE values (see Table S1 of

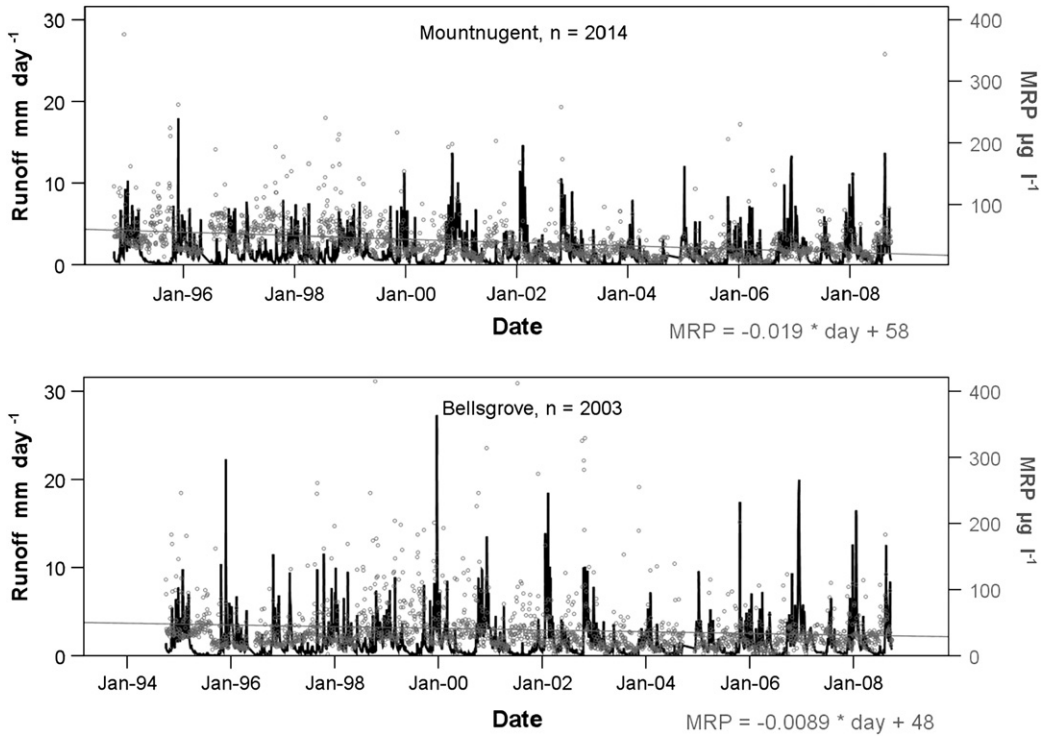


Fig. 3. Empirical runoff (mm day^{-1})and molybdate reactive phosphorus (MRP µg l^{-1}) concentrations (1995–2008) with sample size (n) for the Mountnugent and Bellsgrove rivers. The solid black lines represent measured runoff rates and the hollow grey circles are measured MRP concentrations. The simple linear equation for empirical MRP trend is shown under the x axis.

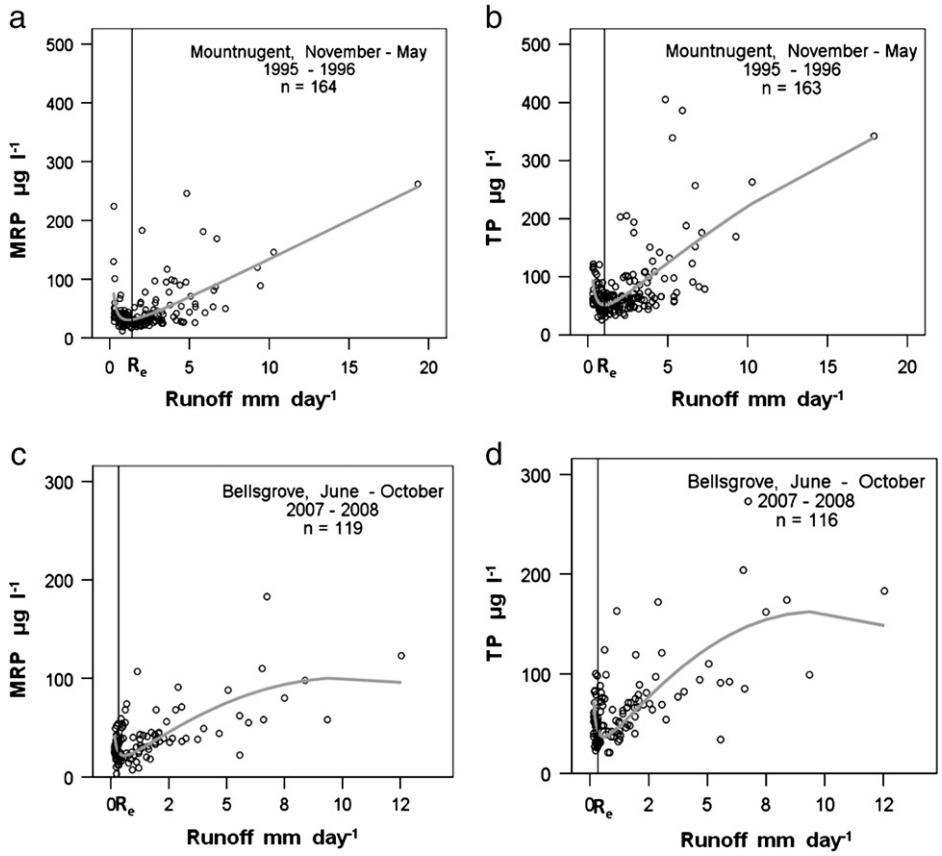


Fig. 4. Examples of calibrated molybdate reactive phosphorus (MRP µg l^{-1}) and total phosphorus (TP µg l^{-1}) seasonal load apportionment models for the Mountnugent and Bellsgrove rivers showing sample size (n).

Supplementary material). The highest validation (78%) and the lowest CV- RMSE (41%) were in Ross winter models. However, these values decreased in summer months to 39% and 52%, respectively. The highest validation in the summer models was Halfcarton (57%). LAM loads were compared with two alternative methods of load estimation: export coefficient and flow-weighted. The relationship between point source MRP loads from LAMs and export coefficients are shown in Fig. S2 of the Supplementary material. The correlation has an overall R^2 of 0.51 ($p<0.01$). However individual subcatchment correlation coefficients range from R^2 of 0.07 to 0.66. The relationships between annual LAM MRP and TP loads with annual flow-weighted loads have R^2 values of 0.95 and 0.97 respectively, highlighting an effective validation of the overall total load estimation (see Figs. S3 and S4 of the Supplementary material).

There was substantial variation between years and between subcatchments in the rates of TP and MRP exports (Table 2). Normalised to catchment area, the highest annual point and diffuse exports of both TP and MRP loads were delivered to the Schoolhouse river. In 1999–2000 the TP diffuse export was 77 kg km^{-2} yr^{-1} of which 79% was in the form of MRP (61 kg $km^{-2}yr^{-1}$). In this year the point source TP and MRP export rates were also higher than in any other subcatchment (TP 27 kg $km^{-2}yr^{-1}$and MRP 13 kg $km^{-2}yr^{-1}$). During the subsequent two year period point source TP and MRP inputs to this river remained high but it was only over these four years that point source exports rates were high as they were not evident for the remainder of the study period so that, for example, over the subsequent four years (2003–2006) the point TP and MRP exports were 4 and 2 kg km^{-2} yr^{-1} respectively. The small Crover subcatchment produced low point source exports of both TP and MRP which in 2003–2004 were only 0.7 and 0.3 kg km^{-2} yr^{-1} for TP and MRP respectively. The large Ross subcatchment produced the lowest diffuse export rates of both TP and MRP with notably low rates in 2005–2006 of 1.8 and 1.2 kg km^{-2} yr^{-1} respectively. During this period diffuse sources were estimated to contribute only 8% and 18% to the MRP and TP exports from this subcatchment.

Time series data were categorised into two distinct stages according to the timing of implementation of P-related POMs in the study catchment: Stage 1 (1995 to 1998, pre-implementation of P mitigation) and Stage 2 (1999 to 2008, post-implementation of P mitigation). Generally, annual diffuse P loads exceeded point loads in terms of magnitude and time duration, except in certain periods from Halfcarton and Ross (Table 2). The magnitude of point P loads in Halfcarton dominated annual P load in 1997–1998 (TP and MRP), 2001–2002 (MRP) and 2003–2004 (TP) and the R_e value showed that point P had flow duration dominance over diffuse loads in 1999–2000, 2001–2002, 2003–2004 (all TP and MRP) and 2003–2004 (TP). In Ross, the magnitude of point P loads was usually higher than diffuse except in 1995–1996 (TP), 1999–2000 (MRP) , 2003–2004 (TP) and 2007–2008 (TP and MRP). However, the R_e values indicated an overall equal but fluctuating flow duration dominance of both P sources (Table 2).

Reductions of point TP (60%, $p<0.05$) and MRP (80%, $p<0.05$) in Mountnugent 1999–2000 are attributed to increased P removal at the Ballyjamesduff WWTP from 1999 (Table 2). The R_e values indicated that the reduction of P delivery to the river in 1999–2000 had a similar affect on MRP and TP point source dominance; point MRP time dominance decreased in winter and increased in summer. However, a second attempt at point P removal (55% point load MRP reduction from 2001–2002, $p<0.05$) in 2003–2004 decreased both seasonal dominance of point MRP and winter TP but had an increasing affect on summer point TP dominance. Attempts of point source P remediation in Ross during 1999–2000 (38% MRP load reduction, $p>0.05$) and 2003–2004 (53% MRP load reduction, $p<0.05$) resulted in the decrease of point P time dominance in both seasons (Table 2). Noticeably, point source TP loads from Ross 2005–2006 dominated temporally (99% TP,

Table 2
Total phosphorus (TP) and molybdate reactive phosphorus (MRP) loads and % seasonal point source duration estimated from load apportionment models.

Diffuse load	Mountnugent		Ross		Bellsgrove		Halfcarton		Schoolhouse		Carrick		Crover	
kg km^{-2}yr^{-1}	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP
95–96	49	24	20	9	40	23	11	5	32	18	34	13	42	25
97–98	39	23	8	4	36	24	9	3	60	45	21	11	22	14
99–00	45	28	12	8	48	36	15	12	77	61	15	9	32	26
01–02	50	25	10	1	55	24	9	2	38	33	29	10	61	29
03–04	40	14	8	3	44	28	4	2	47	30	20	11	33	19
05–06	33	14	2	1	39	17	14	4	29	16	35	11	28	18
07–08	66	36	35	11	57	31	17	7	72	33	50	23	30	16
Point load	**Mountnugent**		**Ross**		**Bellsgrove**		**Halfcarton**		**Schoolhouse**		**Carrick**		**Crover**	
kg km^{-2} yr^{-1}	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP
95–96	14	9	12	9	5	3	9	3	6	4	3	2	9	5
97–98	15	12	16	11	11	7	10	5	9	7	7	5	11	8
99–00	6	2	13	7	15	7	15	5	27	13	4	2	10	6
01–02	8	5	13	9	12	6	7	3	30	10	9	2	3	2
03–04	5	2	7	4	4	2	7	2	4	2	4	1	1	0
05–06	6	2	23	9	8	2	8	2	4	2	7	3	16	5
07–08	7	4	14	7	9	5	5	1	15	9	5	2	13	6
Point time (winter)	**Mountnugent**		**Ross**		**Bellsgrove**		**Halfcarton**		**Schoolhouse**		**Carrick**		**Crover**	
%	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP
95–96	32	42	37	66	17	16	32	22	13	13	7	9	19	15
97–98	27	36	47	42	29	35	51	28	17	16	13	14	20	13
99–00	8	7	32	35	35	11	81	67	51	25	26	22	13	1
01–02	33	54	52	89	43	37	82	85	87	68	58	14	8	4
03–04	5	9	36	68	0	0	91	91	0	0	25	9	1	1
05–06	4	3	99	21	0	0	93	9	24	20	23	21	3	10
07–08	17	18	55	51	28	28	38	21	31	27	41	36	38	29
Point time (summer)	**Mountnugent**		**Ross**		**Bellsgrove**		**Halfcarton**		**Schoolhouse**		**Carrick**		**Crover**	
%	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP	TP	MRP
95–96	4	6	9	8	5	1	22	22	29	31	44	43	24	38
97–98	5	16	45	78	3	4	4	49	4	9	46	59	10	19
99–00	23	17	14	23	20	15	84	48	50	52	26	21	3	16
01–02	1	28	85	97	9	37		92	42	47	33	42	0	13
03–04	12	8	31	55	0	0	68	90	16	0	55	64	10	8
05–06	2	15	100	95	67	16	23	31	15	0	49	11	25	
07–08	28	38	44	43	12	29	59	12	53	53	44	48	47	45

52% MRP) and in terms of load magnitude (92% TP, 88% MRP). In 2003–2004 there was a general large reduction in the time duration dominance of point sourced P, except in Halfcarton (Table 2). These reductions resulted from a number of statistically significant declines in point TP and MRP from the subcatchments.

Accumulated loads (point, diffuse and total) and cattle stocking densities per unit area for each subcatchment during stage 2 are listed in Table 3. Overall, Schoolhouse contributed the greatest load of point TP, diffuse MRP and diffuse TP load, whereas the highest point MRP load was from Ross. Lowest TP and MRP point loads were in Carrick; Ross had the lowest diffuse MRP and Halfcarton the lowest diffuse TP load. The Schoolhouse subcatchment contributed large loads of point and diffuse P during 1999–2000. A similar large peak in diffuse P but not in point P occurred in Crover, 2001–2002. Correlations between point MRP and TP with cattle stocking density from the five rural subcatchments were statistically significant ($R^2 = 0.92$ and $R^2 = 0.9$, $p < 0.05$). Accumulated point MRP and TP loads and accumulated human population densities were positively correlated for the five rural subcatchments ($R^2 = 0.97$ and $R^2 = 0.94$, $p < 0.05$). Furthermore, diffuse TP and MRP were significantly correlated with cattle density ($R^2 = 0.83$, $R^2 = 0.96$, $p < 0.05$) after exclusion of data from the Ross sampling point.

Extreme runoff events contributed a large proportion of annual P load to rivers. In 2007–2008, runoff greater than 10 mm day^{-1} delivered 35% of annual loads over 20 days to Mountnugent. Phosphorus source and trend were altered in 2007–2008 (Table 2). Diffuse P loads from

Table 3
Accumulated total phosphorus (TP) and molybdate reactive phosphorus (MRP) loads for the P mitigation stage (1998–2008) from load apportionment models. Cattle and human population densities are also shown.

River	Point MRP kg km^{-2}	Point TP kg km^{-2}	Diffuse MRP kg km^{-2}	Diffuse TP kg km^{-2}	Total MRP kg km^{-2}	Total TP kg km^{-2}	Cattle LU[a] km^{-2}	Rural human km^{-2}
Bellsgrove	43	95	272	486	316	581	1094	98
Carrick	21	60	127	299	148	358	1050	86
Crover	38	84	215	368	253	452	1078	94
Halfcarton	26	82	55	119	81	201	892	78
Mountnugent	32	62	234	468	266	530	1381	148
Ross	73	139	50	133	123	272	1207	66
Schoolhouse	70	161	344	528	414	688	1531	134

[a] Livestock unit.

Bellsgrove, Mountnugent, Ross and Schoolhouse had declined substantially from 1999–2000 to 2005–2006. However, the high P load in 2007–2008 altered the 10-year downward trend and led to an overall increase over the time period represented. The total rainfall over the two hydrological year periods of 2007–2008 and 1999–2000 were similar (2264 and 2224 mm, respectively) but there were more extreme flows ($\geq$10 mm day^{-1}) in each subcatchment in 2007–2008 which delivered larger proportions of the annual P load.

There were also point P load increases for Mountnugent ($p>0.05$), Bellsgrove and Schoolhouse ($p<0.05$) in 2007–2008, which also altered the declining trend of preceding years. Conversely, point P decreased ($p>0.05$) in Halfcarton and Ross ($p>0.05$) during the same period.

4. Discussion

4.1. Point sources

A priori, there was an expectation of a large point P source from the subcatchments with urban P loading; this was supported by the results. However, the results also provide evidence of a point source influence in rural subcatchments, and this is in-line with findings from previous studies in which point loadings can be attributed to combinations of P from cattle access points (Sharpley and Syers, 1979; Jarvie et al., 2010), farmyard leaks (Edwards et al., 2008; Edwards and Hooda, 2008), artificial land drains (Kurz, 2000; Withers and Jarvie, 2008) and septic tanks (Arnscheidt et al., 2007; Jarvie et al., 2010). In the rural areas of the Sheelin catchment, septic tank systems are the main method of disposal of human waste. Typically, the effluent from the tanks is managed to drain to a soakaway in which the nutrients are dissipated into the soil (Harper, 1992). Septic tank P transfers were considered to contribute minimal P to surface waters owing to higher (than WWTP) P retention in soils. Carvalho et al. (2005) and LLCMMS (2000), however, showed that P loads from septic tank systems to water bodies may be higher than expected, owing to poor design, maintenance or position, the piping of wastewater to a nearby watercourse and possibly exceedence of dilution in receiving waters due to cumulative loads.

Ross showed the largest accumulated point MRP loading per unit area even following implementation of P-mitigation (at WWTP). Ross has a relatively low rural population density, a lower urban population than Mountnugent and the highest MRP:TP ratio (0.53). A high proportion of soluble P contained in TP indicates that the WWTP is a dominant contributor of MRP. In addition to source load magnitude, the estimation of R_e values showed that temporal dominance is also an important issue. For example in summer 2001–2002, point TP and MRP contributed, respectively, 49% and 57% to annual P loads but the R_e value indicates that point loads dominated annual flows (85% TP and 97% MRP) (Table 2). This indicates that the river was at a much higher risk to eutrophication in the ecologically sensitive summer period than indicated by point P levels alone.

Mountnugent had the second highest MRP:TP ratio (0.51), which was expected because of WWTP loading. However, the overall magnitude of point P contribution in Mountnugent was relatively low (Table 3). The seasonal R_e values, in contrast with Ross, indicated that point source magnitude was more influential than time duration in both winter and summer models. Therefore, the time dominance of diffuse loads indicated that although there is point P pressure in the subcatchment, catchment managers should focus more resources on reducing diffuse P inputs. The subcatchments, Bellsgrove and Schoolhouse, contributed some of the greatest point MRP and TP load from annual load which indicates the presence of constant sources of P in these rural subcatchments (Table 3).

Schoolhouse had the highest cattle stocking density (Table 3). A significant correlation of point P load with cattle stocking density implies that there is a positive linear relationship between the two factors, and may reflect the direct delivery of livestock manure, farmyard runoff and slurry to the rivers (Jarvie et al., 2010). Rural human population densities and point MRP and TP were also positively related, and this shows that malfunctioning septic tanks may be a source of direct, localised discharge into rivers (Jarvie et al., 2010). However, the relationships were non-significant ($p>0.05$) when the rural populations of Mountnugent and Ross were included. This can be attributed to the impacts of loadings from WWTP and industrial P in Ross and Mountnugent and reflects the changing P loading dynamics from rural to urban.

4.2. Diffuse sources

Soils in the Crover and Mountnugent subcatchments are predominately poorly drained gleys. Well drained brown earths cover the Ross, Halfcarton and Schoolhouse subcatchments, whereas Bellsgrove and Carrick have a mixture of both soil types. Previous studies have found that poorly drained soils have the highest Q5:Q95 ratios and can transfer the most diffuse P load (Jordan et al., 2005a) and this is evident from data presented here. The overall highest accumulated diffuse P loads were from the Bellsgrove, Crover, Mountnugent and Schoolhouse subcatchments (Table 3). In the rural subcatchments, transfers of diffuse MRP and TP are positively correlated with cattle stocking densities (Table 3). However, Tunney et al. (2007) found that grazing animals significantly influence TP, and not MRP, concentration in overland flow: Douglas et al. (2007) also reported a dominance of particulate P from drumlin soils under grassland. Diffuse TP and MRP loads were positively related to rural human population densities, indicating the potential contribution of diffuse P inputs from septic tank soak-aways. Halfcarton, which has a similar soil type to Schoolhouse, has lower agricultural intensity, fertiliser application rates and cattle stocking densities, and contributed the lowest TP diffuse loads. This suggests that excessive P loading is more influential than soil desorption risk and a threshold for P transfer exists between P loading and soil type.

Previous studies have shown that runoff from grasslands generally carries little sediment and that the TP load is mainly dominated by the soluble P form (Heathwaite et al., 1997). Molybdate reactive P from grasslands comes from P release from recently applied, inorganic fertilisers that remain unincorporated (Torbert et al., 1999, Withers et al., 2001); grasslands that are fertilised lose more MRP than unfertilised lands (Kleinman et al., 2002). The rapid increase of P load delivery during extreme rainfall events in 2007–2008 caused the acceleration of eutrophication in Lough Sheelin in 2008 (data not shown). Increases in P transfers synonymous with extensive runoff have been reported elsewhere (Webb et al., 1997; Tunney et al., 2000; Shigaki et al., 2007). Diffuse P was predominately in the form of MRP, which, as the most bioavailable form of P, has been shown to have a large impact on the ecological status of a lake (Foy et al., 1995).

Therefore, diffuse sources of P are most likely owing to high cattle stocking densities, and the extents of farmed land overall (84%), particularly fertilised pasture (51%), and septic tank soak-aways. However, agricultural sources are usually the main contributor of diffuse P (Jennings et al., 2003) and a greater focus on measures targeting land fertilisation and slurry spreading, such as those being reviewed in the Nitrates Directive National Action Programme (Fealy et al., 2010), would facilitate the recovery of Lough Sheelin.

4.3. P mitigation appraisal

Variations in P export rates were recorded during the mitigation stage in all subcatchments, although levels of MRP and TP declined overall (Fig. 3). The significant decline in point P in Mountnugent and Ross reflected the increased P removal at WWTPs and restrictions on industrial discharges. However, despite the large point decreases in Mountnugent 1999–2000, there was an increase of point MRP the following year (Table 2). This increase may reflect the fact that several sources of point P still existed in the catchment. Nevertheless, point

MRP declined in 2003–2004, despite a rapidly growing catchment population (30%). The decline coincided with the implementation of septic tank and agricultural bye-laws. Point P also declined in 2003–2004 in the rural subcatchments that underwent the implementation of both byelaws (Bellsgrove, Carrick, Crover and Schoolhouse). However, extreme runoff events during 2007–2008 reversed these decreases (Table 2). Halfcarton and Ross, the subcatchments with lower Q5:Q95 ratios, free draining soils and lower P desorption indices, are less susceptible to high surface runoff (Table 1). The R_e values support this evidence; point sources in Ross dominate both load magnitude and flow duration owing to the smaller influence of diffuse sources (Table 2). However, diffuse sources usually have a domineering impact in the gley soil subcatchments (Table 2). A decrease in point P loads in Halfcarton and Ross during 2007–2008, as opposed to increases in the other subcatchments, is indicative of a link between runoff and the magnitude of P load from point sources where soils are frequently water-logged.

In Ross, the R_e value showed that point P loads generally dominated flows during the 5 month summer seasons (Table 2). Ross does not contribute excessive P loadings to the annual catchment P budget but high concentrations and dominance of point MRP during the summer have a localised affect on eutrophication in the river (e.g. Jordan et al., 2005b). The installation of temporary P removal in 2003–2004, and the control of industrial discharges, reduced the percentage of times that summer flows were dominated by point P, thereby reducing the river eutrophication risk. However, the point P increase in 2005–2006 is attributed to a large accidental spillage from the WWTP during August 2005 (Kerins et al., 2007) and the management impact of P mitigation is not apparent.

Lough Sheelin was reclassified from mesotrophic to eutrophic in 2008. A policy implemented in 2003 to reduce diffuse P loads restricted land spreading of slurry after the end of September of each year. Excessive rainfall occurred in June–July 2007 and large surface runoff rates incorporated newly applied fertilizers and slurry that went on to have a negative impact on the ecological status of Lough Sheelin.

4.4. Model strengths and weaknesses

The highly significant relationship between LAM annual P loads and flow-weighted loads ($R^2>0.98$, $p<0.01$) demonstrates that the LAM accurately describes P loads. Flow-weighting is a standard and acceptable method of nutrient load estimation using empirical data (Salles et al., 2008) and the LAM method improves on this by separating the P load into sources. The fact that the LAM MRP estimates agree closely with results of the flow weighted method but poorly with export coefficient estimations of point MRP load indicates possible inaccuracy in the export coefficient method. The methods used to estimate urban and rural point P loading for export coefficient analysis have many potential sources of error: septic tank and WWTP effluents vary temporally (Jordan et al., 2005a; Neal et al., 2005), the interpolation of rural population levels is susceptible to error (Reibel and Agrawal, 2007) and published export coefficients for septic tanks are variable (Carvalho et al., 2005). Subcatchment specific information is crucial for accurate calculations of export coefficients. The method of using direct river measurements to develop and calibrate the LAM has the advantage of deriving P load estimates that incorporate known and unknown origins of P in the catchment.

The general relationships that describe point and diffuse sources of P delivery with flow (Manczak and Florczyk, 1970; Chapman, 1996; Edwards and Withers, 2007; Jordan et al., 2007; Neal et al., 2010) used in LAM described here are also applied in other LAMs (Mohaupt 1986; Lennox et al., 1997; Albek, 2003; Bowes et al., 2008). However, compared with existing LAMs, the approach adopted in the current paper is the best fit for the study database. An increased explanation of variation of P with discharge was obtained by calibrating models with seasonal data. The unconstrained model coefficients allowed for the consideration of concentration-discharge hysteresis which was observed in most model outputs.

The amount of variation observed in model plots and validation percentages can be influenced by model theoretical assumptions and sampling or analytical processing. The assumptions made in the conceptualisation of the LAM model may have implications for model outputs. The point-sourced LAM estimations may include the effects of riverine processes on P (Dorioz et al., 1998; Bowes et al., 2003, 2005b, 2008; House, 2003; Edwards and Withers, 2007; Stutter and Lumsdon, 2008) and the infrequent occurrence of pollutant events (Jordan et al., 2007). Nevertheless, LAM estimations of source loads from the TP models follow a similar trend to MRP, which confirms that the relationships exist and sorption of MRP does not have a significant influence on the results.

Inaccuracies in sampling, analysis, data quality assurance and data manipulation can have an impact on modelled outputs. Left-censored data that were substituted with half the detection limit had a negligible impact on model outputs owing to the percentage of data censored (less than 1%), the large sample size and a single limit of detection (Zhang et al., 2004). Despite the high frequency of sampling, high flow events remain relatively under-represented in the current study. Low to medium flow occurred more frequently in the study dataset resulting in a possible quantitative bias towards point sourced loads. However, the unconstrained term for exponential P delivery to the river is necessary, owing to hydrochemical hysteresis and the high percentage of P load delivered to the rivers via this pathway (Bowes et al., 2005b; Haygarth et al., 2005; Edwards and Withers, 2008). Ensuring that water quality is sampled across a complete range of flows is necessary to meet the data needs of LAMs.

The use of runoff in the LAM presented here limits its application to rivers in catchments with a spatially uniform dominance of surface and near-surface pathways. However, normalising flow for catchment area gives catchment managers important information on areas of higher P export risk (Vagstad et al., 2004; Bechmann and Stålnacke, 2005). The relationships and assumptions of the LAM components also restrict model application to catchments where P delivery requires source apportionment. Therefore, the LAM developed here using data from rivers in the Sheelin catchment, which contains surface pathway dominated rivers and a high risk of diffuse and point source P transfers – the major catchment typology in Ireland – is highly applicable elsewhere. Despite many possible confounding factors, the relationships conceptualised in LAMs and reported by Jordan et al. (2007) are validated here.

5. Conclusions

Managers of catchments where eutrophication pressures are regarded as high require accurate knowledge of the status of P, source of P and the interactions of P with the local environment in order to implement mitigation policy effectively. LAMs provide a solution to two important problems concerning the mitigation of nutrient-induced aquatic impacts: the approximation of P from various sources in terms of magnitude and duration, and the need for a feasible, cost-effective monitoring programme that provides information on actual environmental processes with acceptable levels of statistical certainty. The LAM developed in this research, by estimating point and diffuse source loads and providing an insight into P dynamics, provides a means of supporting decision making relating to catchment management. The models, which are generally applicable, have been applied at subcatchment level and provide a means of evaluating the effectiveness of POMs.

Subcatchments contributing disproportionately high loads of P to annual loads were identified in the current research. The detection of point P transfers to rural subcatchments identified the risk, especially to river status, of sources that may not have been considered a problem previously. The relationships of point and diffuse P loads with rural population densities confirm that septic tanks can contribute P to rivers

via the response to rainfall and by direct discharge. Large transfers of P in extreme flows in 2007–2008 are likely to have caused a decline in the ecological status of Lough Sheelin. A seasonality component was evident in all river data: timing and delivery of P during both summer and winter seasons are crucial to the effective management of P.

Temporal and spatial heterogeneity of P concentration and source load existed within and across the subcatchments as a result of differences in environmental attributes and P loading. The effectiveness of P mitigation was largely influenced by the interaction of these environmental attributes, P loading and the extent of implementation of POMs at a local level. Nevertheless, significant temporal changes in P source loading as a result of P mitigation (WWTP increased P removal, septic tank bye-laws, industrial discharge licensing and agricultural bye-laws) were detected by the LAMs. These results imply that, to a certain extent, the policies introduced were effective in reducing P. However, the results of some remediation attempts were masked by process such as high rainfall and accidental P transfers. Thus events outside the control of mitigation management have attenuated a signal of improving water quality over short time-periods, a finding that highlights the need for long-term monitoring that fully samples the flow range in rivers, especially in catchments with multiple source pressures (e.g. point and diffuse P) and multiple impact receptors (rivers and lakes).

LAMs have been used in this study to identify subcatchments of higher P export risk and sensitive transport pathways as required by the WFD. In addition to the monitoring point and diffuse load magnitude, the frequency of occurrence of P source during the year is an important factor. The estimation of the R_e parameter is vital to the use of LAMs in the implementation of measures to control eutrophication in rivers, and therefore LAMs deserve serious consideration as tools to facilitate evaluation of the impact of different P mitigation strategies at both sub-catchment and full catchment scales.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.scitotenv.2011.02.016.

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